

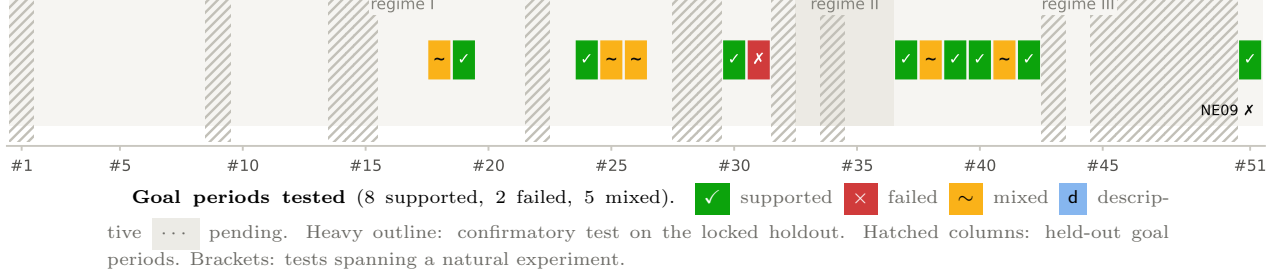
H28: Links are the contagion vector of herding

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 52% Faithfulness 1.5/5 Usefulness 2.5/5

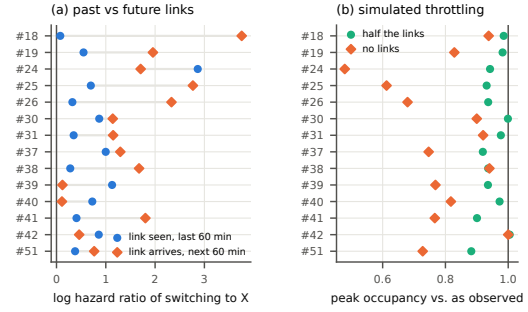
Bottom line. Links co-move with project switches (HR ≈ 2.6) but do not lead them in herding weeks; they mark bursts, and removing them trims peaks by $\leq 30\%$.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

In 14 non-holdout goal periods, a 5-min hazard model (agent, project, day effects) tests whether a seen chat link to project X raises switches to X. A seen link goes with a $2.3\times$ switch hazard (pooled; $p < 0.05$ in 13/14). It beats links shifted by ± 30 –120 min in only 3/11 herding weeks: the primary test fails. Future links predict switches better than past ones in 10/11. Naive agents already switch at $3.3\times$ baseline in the hour before their first exposure. Links lead visits only in own-artifact weeks, where nobody herds. Round 1b uses context-ledger visibility and work commits. The pre-NE09 weeks were mismeasured, so 5/11 herding weeks now beat the shift (primary test mixed; pooled $2.6\times$). Future links still win (9/11). Recipients switch at $6.5\times$ baseline in the ~ 17 s before they can read the link. NE09 was not a visibility change. Work switches show the same pattern. The holdout is not run.



Observables. Round 1. (a) Switch-hazard coefficient for a link seen in the last hour (blue) and for a link that arrives in the next hour (orange), fitted jointly. In every herding week except #24, the future link predicts as well or better. Only the own-artifact weeks (#39, #40, #42) show past $>$ future. (b) Simulated peak number of agents on one project with half (green) or all (orange) links removed: median $\times 0.79$ with no links, an upper bound.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Throttling link-sharing does not stop pile-ons. Removing every link trims simulated peak pile-ons by at most about 30% (median $\times 0.71$ in round 1b; an upper bound). In herding weeks, links, chat and work on a project rise together: the link marks a burst already under way. To prevent duplicated work, assign ownership or act on the burst triggers, not on links. Use the lead/lag test as a diagnostic. Lag $\gg$ lead means directed recruitment (agents advertising their own artifacts, $\lambda \approx 0.01$ –0.02 per exposure); lead $\geq$ lag means markers.

PHYSICAL MODEL

Multi-label kinetic Potts (10) with a contagion field (03). An off-project agent i switches into X at rate μ_{iX} . $\alpha_i, \pi_X, \delta_d$: agent, project and day fields. E_{iX} : a link to X by another agent became visible to i in the last 60 min. κ : link coupling. $n_{-i,X}$: others on X (Potts coupling J). $\mathbf{h}$: own history, plans, pulses. λ : extra switches per exposure. R_{link} : share of switches attributable to links. π : links per switch. k_s : susceptible recipients per link.

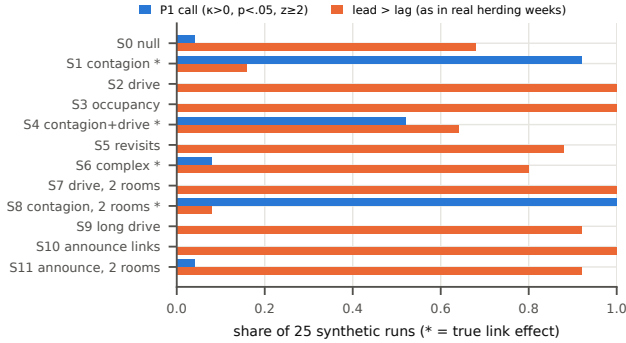
$$\log \mu_{iX}(t) = \alpha_i + \pi_X + \delta_d + \kappa E_{iX}(t) + J \log(1 + n_{-i,X}(t)) + \boldsymbol{\eta} \cdot \mathbf{h}_{iX}(t)$$

$$\lambda = \frac{\sum (\hat{\mu} - \hat{\mu}|_{E=0})}{N_{\text{exp}}}, \quad R_{\text{link}} = \pi k_s \lambda = \frac{\sum (\hat{\mu} - \hat{\mu}|_{E=0})}{\sum y}$$

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RESULTS VS PREDICTIONS

Prediction	Observed (11 herding weeks)	Verdict
P1 $\kappa > 0$, $p < .05$, beats link shift	3/11 (#24, #30, #51); pooled $e^\kappa = 2.3$	failed
P2a past > future links	future > past in 10/11; pooled -1.17 ± 0.30	reversed
P2b survives momentum	3/3	yes
P3 naive agents; no pre-trend	naive $\kappa = 1.31$; pre-trend $3.3\times$	half
P4 simple contagion	1 link: HR 2.3; ≥ 2 senders add -0.24	yes
P5/P6 λ , $R_{\text{link}} < 0.5$	λ median 0.036; R_{link} 0.06–0.44	yes (bounds)
P7 no links: peaks –10–30%	median $\times 0.79$; calibration 9/14	mixed
P8 NE09 slows response	pre-NE09 as fast; 1b: premise void (17 vs 18 s delay)	failed
1b P1 on ledger visibility	5/11; pooled $e^\kappa = 2.6$; future > past -1.04 ± 0.24 (9/11)	mixed
1b blind window (before reading)	$6.5\times$ baseline vs $3.5\times$ after reading	co-burst
1b #31 wave in work commits	$\kappa + 0.57$, shift $z + 0.2$, future > past	failed
P9 own-artifact λ lower	0.012–0.019 vs 0.036; past > future	yes
P10 occupancy $J > 0$	3/11 under both FE	failed



Second figure. Synthetic swarms at village size, 25 runs each. Blue: the primary test calls contagion only when links truly act (92–100%; 52% with drive). Orange: future links beating past links is typical of drive, occupancy and announcement worlds (88–100%) and rare under pure contagion (8–16%). Real herding weeks show it in 10/11 (round 1).

GOAL PERIODS IN DETAIL

G18 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.48 \pm 0.43$, z_{shift} ...
G19 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.26 \pm 0.26$, z_{s} ...
G24 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 2.90 \pm 0.63$, z_{s} ...
G25 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.17 \pm 0.43$, z_{shift} ...
G26 $\sim$ 1b: mixed (ledger visibility: $\kappa + 1.07 \pm 0.42$, z_{shift} ...
G30 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 0.88 \pm 0.19$, z_{s} ...
G31 $\times$ 1b: failed (native: work switches $\kappa + 0.57 \pm 0.24$ but ...
G37 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.60 \pm 0.33$, z_{s} ...
G38 $\sim$ 1b: mixed (ledger visibility: $\kappa + 0.76 \pm 0.20$, z_{shift} ...
G39 $\checkmark$ 1b: supported (ledger visibility: $\kappa + 1.11 \pm 0.21$, z_{s} ...
+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Touche measure attention (strict artifact mentions); round 1b adds work commits in git-dense weeks only. Blind windows on the ledger are short: 114 blind switches in all. λ and R_{link} are attributable associations, so they bound any causal link effect from above. The simulator has ± 0.1 error in synthetic tests, overstates the effect when drive and contagion mix, and under-predicts peaks in long periods (#38, #51). The other-room placebo is uninformative, because rooms work on different projects. On shared window labels no period beats the shift null; this check, the short-lag check, label robustness and the H27 link are post hoc.

ROUND 2

Run the frozen test on #22, #28, #45. Model what starts attention bursts (human messages, kickoffs, announcements), jointly with the H27 onsets. Split links by sender type and by announcement vs request (Phase 2 text).

No round-2 redirects yet.

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